

DECSI PLSA — Technical Specification

Field	Value
Document	Technical Specification (v2.0)
Product	PLSA — Prize-Linked Savings & Customer Engagement Platform
Institution	Dedebit Credit and Saving Institution (DECSI)
Vendor	Seqela Technologies
Date	2026-07-28
Status	Aligned to DECSI source documents
Source inputs	<i>PLSA For DECSI</i> (Seqela proposal); <i>PLSA Project Marketing Plan & Strategy</i> (June 2026)

1. Purpose and product positioning

1.1 What PLSA is for DECSI

PLSA is **not only** a classic prize-linked savings account. Per DECSI's approved marketing direction, it is DECSI's **premier customer engagement, loyalty, and promotional platform** — a growth engine that connects customers to the full DECSI ecosystem through:

- savings incentives and prize draws
- digital adoption (especially **AdoPay**)
- cross-selling (loans, merchant payments, School ERP, SACCO / multipurpose)
- financial education and community engagement
- referral and loyalty rewards

Classic PLSA mechanics (entries from savings activity; **principal never at risk**) remain the core of the draw model. The platform extends entries to other positive behaviours DECSI wants to grow.

1.2 Strategic vision

Establish PLSA as DECSI's primary engagement and rewards platform that drives financial inclusion, savings culture, digital financial services, and long-term customer relationships.

1.3 Institutional context (scale assumptions)

Factor	Value (from DECSI proposal)
Clients	1,000,000+
Branches	200+ (including 4 in Addis Ababa)
Lending scale	> ETB 10 billion disbursement from 2024
Digital channel	AdoPay mobile banking
Geography	National MFI footprint; strong Tigray / Mekelle base

The platform must be designed for **millions of accounts** and concurrent national campaigns.

1.4 Source-document alignment

Document	Role in this spec
PLSA For DECSI	Product model, Campaign → Phase → Draw, segments, economics, architecture, compliance, rollout
PLSA Marketing Plan & Strategy	Five marketing pillars, multi-behaviour rewards, scholarship prizes, channels, KPIs, ecosystem products

2. Business objectives

Mapped from both source documents.

ID	Objective	Platform implication
O1	Strengthen voluntary (and compulsory) savings mobilization	Ticket rules on deposits, balances, frequency
O2	Expand financial inclusion (rural, women, youth, underage, PwD)	Segment campaigns, accessible channels (SMS, branch, agent)
O3	Increase customer acquisition	Referral entries; campaign tagging of new accounts
O4	Promote AdoPay and digital DFS	Bonus entries / multipliers for digital deposits & transactions
O5	Increase engagement & retention	Continuous Campaign → Phase → Draw cycles
O6	Increase institutional revenue & sustainability	Deposit lift → lending capacity; fee/txn revenue analytics
O7	Strengthen brand & public trust	Transparent draws, public winner announcement, audit packs
O8	Cross-sell DECSI products & Seqela ecosystem	Entry rules for loans, repayments, School ERP, SACCO, merchants

3. Product model

3.1 Behavioural PLSA core

- Customers keep **full access** to savings (no lock-box requirement for participation).
- Each qualifying action increases **probability of winning** (more tickets).
- Prize funds come from **institutional marketing / incentive budget**, **not** from deducting customer deposits.
- Interest on deposits continues under DECSI / NBE deposit rules.

3.2 Campaign → Phase → Draw hierarchy

This is the **canonical structure** for DECSI PLSA.

Campaign	National or segment initiative (may run in parallel)
└─ Phase	Time-boxed stage (e.g. monthly prizes → regional → grand)
└─ Draw	Weekly / monthly / quarterly / annual execution event

Example campaigns

- National Savings Campaign
- Women Financial Empowerment Campaign
- Youth Digital Savings Campaign

- Harvest / Farmer Seasonal Campaign
- Educational Scholarship Campaign (“Save Today, Educate Tomorrow”)

Example phases within a campaign

1. Phase 1 — Monthly prizes
2. Phase 2 — Regional prizes
3. Phase 3 — National grand prizes

Draw cadence (configurable per phase)

- Weekly, monthly, quarterly, annually

Multiple campaigns may run **simultaneously** with independent eligibility and ticket pools.

3.3 How customers earn entries (tickets)

From the marketing plan **Pillar 5** and DECSI proposal eligibility model.

Behaviour	Typical entry effect	Notes
Open / maintain savings account	Base enrolment + ongoing eligibility	Regular, voluntary, compulsory, fixed-term, group
Increase savings balance / deposit amount	Primary ticket generator	Probability rises with amount & frequency
Regular monthly deposits	Consistency bonus	Encourages habit
Use AdoPay (deposit / transfer / pay)	Digital multiplier or bonus tickets	Strategic digital push
Timely loan repayment	Loyalty tickets	Cross-sell / retention
Refer new customers	Referral tickets	Acquisition
Active account status	Soft eligibility gate	Dormant = ineligible
Use DECSI School ERP / scholarship path	Education-campaign tickets	Ecosystem
DECSI SACCO / multipurpose cooperative activity	Group / partner tickets	Ecosystem
Merchant / QR / bill payments (AdoPay)	Digital engagement tickets	Optional rule packs

Ticket formula (configurable per campaign/phase)

```

tickets = f(deposit_amount, deposit_frequency, balance_basis, channel_multiplier,
behaviour_bonuses)
tickets = min(tickets, max_tickets_cap)
if not eligible: tickets = 0

```

Recommended Phase-1 default for pure savings phases:

- Unit: e.g. every **ETB 500** net qualifying deposit (or ADB unit) = 1 ticket
- Cap per customer per draw period (anti-whale)
- **AdoPay multiplier** e.g. 1.25×–2× for digital-channel deposits

Exact unit, caps, and multipliers are Board / Marketing parameters (see §17).

3.4 Prize catalogue

Cash and in-kind (from DECSI proposal)

Tier	Timing	Example prizes
Monthly	Frequent, many winners	Cash, smartphones, motorcycles, appliances, solar kits
Quarterly	Higher value	Agri equipment, business startup packages, livestock, shop equipment
Annual grand	Flagship ceremony	Houses, vehicles, tractors, large capital packages

Segment-tuned prizes

Segment	Prize examples
Women	Business equipment, household assets, micro-enterprise capital
Youth	Smartphones, laptops, scholarships, startup grants
Underage / students	School supplies, tablets, education support funds
Rural farmers	Farming equipment, irrigation tools, livestock/inputs
Persons with disabilities	Mobility equipment, assistive tech, small business grants

Educational Scholarship Rewards (marketing plan)

- 1–5 year scholarship packages for winners' children
- Coverage: KG → primary → secondary → preparatory (to Grade 12)
- At approved high-performing schools in **Tigray Region** (extendable)
- Fulfilment is a multi-year obligation tracked in Prize Management (not a one-time cash post)

3.5 Products promoted through PLSA

Savings: regular, voluntary, compulsory, fixed-term, group

Credit: individual, agricultural, SME, women's, youth, emergency/consumption

Digital: AdoPay, digital payments, merchant / QR, transfers, bill pay

Additional: financial literacy, BDS, community financing, School ERP, partner / SACCO services

4. Target segments and campaign packs

Segment	Strategy hooks	Channel bias
Existing DECSI customers	Upsell balances, AdoPay, multi-product	Branch + SMS + AdoPay
Prospects / unbanked	Acquisition campaigns, community events	Field + radio + branch
Women	Dedicated campaigns, women's associations	Group + literacy workshops
Youth / startups	Digital-first, campus, challenges	AdoPay + social
Underage (children/students)	Parent-linked accounts, education prizes	Branch + schools
Farmers	Harvest-season campaigns, cooperatives	Rural branch + radio (Tigrigna)
PwD	Inclusive onboarding, association outreach	Accessible branch + SMS
MSEs / merchants	Merchant payments, business prizes	AdoPay merchant
Cooperatives / groups	Collective participation recognition	Coop meetings
Education	School ERP + scholarship narrative	Schools / Seqela ERP

Platform must support **segment tags** on campaigns, enrolments, and analytics.

5. Prize pool economics (for system configuration)

Model	Prize pool as % of PLSA-mobilized deposits
Conservative	0.5%
Balanced (recommended initial)	1%
Aggressive	2%

Example (moderate participation): ETB 1.5bn annual deposits → **ETB 15m** annual prize pool at 1%.

Deposit lift scenarios (assumptions from proposal)

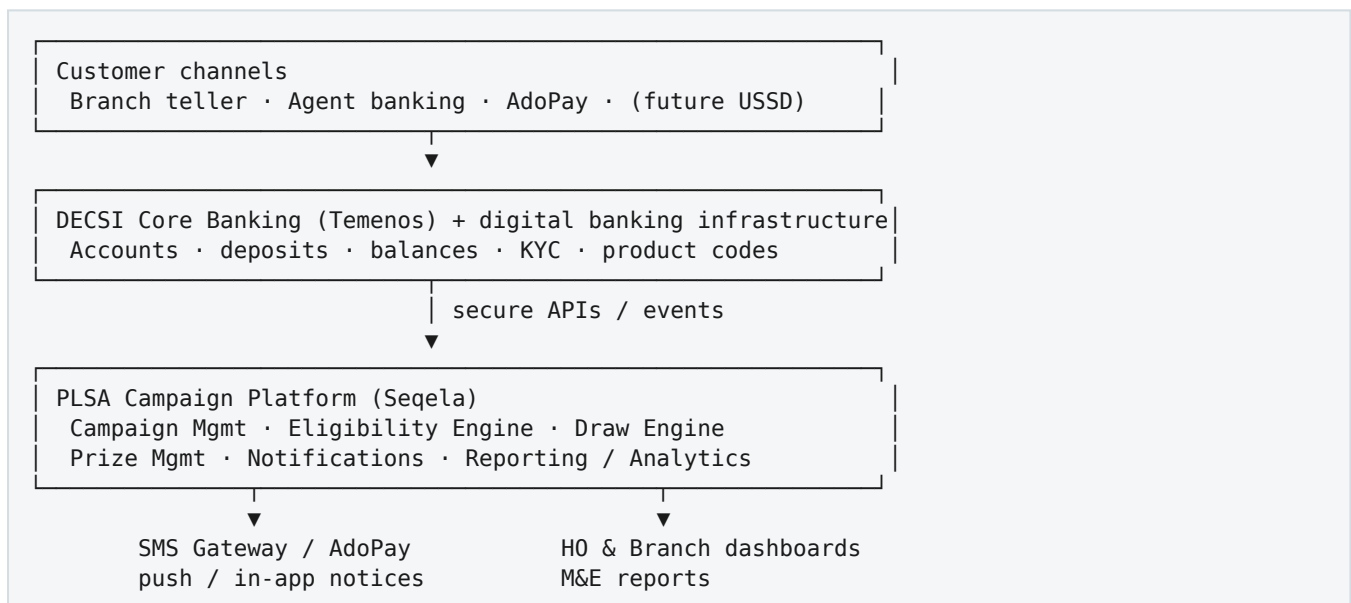
Scenario	Participation	Participants	Monthly deposits	Annual deposits
Conservative	10%	100,000	ETB 50m	ETB 600m
Moderate	25%	250,000	ETB 125m	ETB 1.5bn
High growth	40%	400,000	ETB 200m	ETB 2.4bn

Assumption: average **ETB 500 / month** per participant.

System requirement: Campaign budget module must enforce prize inventory + cash budget ≤ approved pool; alert when remaining pool < threshold.

6. High-level architecture

Aligned to *PLSA For DECSI* §11.



Principle: PLSA is an **integrated service layer** — it does **not** replace CBS, AdoPay, teller, or agent systems.

6.1 System of record

Data	SoR
Customer, accounts, balances, deposits, interest	Core banking
Campaigns, phases, rules, tickets, draws, winners, prize inventory/fulfilment	PLSA platform
AdoPay session / payment UX	AdoPay
SMS delivery	DECSI SMS gateway

6.2 Design principles

Interoperability · Scalability (1M+ clients) · Security · Automation · Transparency / auditability

7. Core system components (functional spec)

7.1 Core Banking Integration Layer

Functions

- Retrieve customer / account master data
- Monitor savings deposits (near-real-time event or frequent batch)
- Track balances for ADB / eligibility
- Verify account status and product eligibility
- Post cash prize credits / GL prize expense (idempotent)

Triggers: When a qualifying deposit (or other mapped txn) posts in CBS → evaluate eligibility → grant tickets.

Channels feeding CBS (hence PLSA): branch teller, agent banking, AdoPay.

7.2 Campaign Management System

Admin (HO) configures:

- Campaign name, status, duration, segment targeting
- Phases and draw schedule
- Eligible account / product types
- Geographic scope (branch / district / region / national)
- Min deposit / balance rules
- Ticket formula, caps, channel multipliers
- Prize categories and inventory assignment
- T&Cs version

Controls: Maker–checker before go-live; pause/close; simultaneous campaigns.

7.3 Eligibility Engine

Continuously (or on txn event) evaluates:

- Deposit amount thresholds
- Deposit frequency
- Account type / segment (youth, women, voluntary, etc.)
- Branch geography
- Phase membership and freeze windows
- Behaviour bonuses (AdoPay, referral, repayment, ERP/SACCO flags)

- AML/KYC hold, staff exclusion, dormancy

Outputs immutable **TicketGrant** records with input hash for audit.

7.4 Draw Engine

- Automated schedule + manual dual-control execute
- Secure RNG; proportional tickets; no replacement within tier (configurable)
- Pre-draw eligibility re-check
- Fraud controls (duplicate identity, velocity)
- Full audit log (census checksum, algorithm version, seed commit/reveal, winners)
- Optional independent observer attestation

7.5 Prize Management System

- Register prizes (cash amount or in-kind SKU; scholarship multi-year package)
- Assign to campaign/phase/tier
- Inventory tracking
- Winner claim → verify → fulfil (cash credit / in-kind handover / scholarship enrolment)
- Status: PENDING_CLAIM | VERIFIED | PAID | DELIVERED | IN_PROGRESS_SCHOLARSHIP | FORFEITED | REJECTED

7.6 Notification and Communication System

Integrate **DECSI SMS gateway** and **AdoPay notifications**:

- Participation confirmation
- Ticket / entry updates
- Draw reminders
- Winner announcements
- Promotional / literacy messages
- Deposit nudges

Languages: Amharic, **Tigrigna**, English (template packs). Radio winner scripts reuse same result feed.

7.7 Reporting and Analytics Dashboard

Real-time / near-real-time:

- Participating customers
- Deposit growth trends
- Campaign performance by branch / region
- Prize distribution stats
- Geographic participation
- AdoPay usage among PLSA participants
- Cross-sell funnel metrics (loan apps, multi-product ownership)

Report levels: branch · regional · national

8. Domain model (logical)

```
Campaign
  phases[], segment_tags[], geo_scope, status, terms_version, budget_pool

Phase
  campaign_id, sequence, start/end, draw_cadence, ticket_rules_json

Draw
  phase_id, scheduled_at, freeze_at, status, census_checksum,
  algorithm_version, seed_commit, seed_reveal, authorized_by[]

Enrolment
  campaign_id, customer_number, account_number, branch_id,
  segment_tags[], channel, opted_in_at, status

TicketGrant
  draw_or_period_id, enrolment_id, tickets, basis_metrics,
  formula_version, input_hash, source_txn_refs[]

PrizeItem
  type (CASH|INKIND|SCHOLARSHIP), value_or_sku, years?, inventory_qty

PrizeAssignment
  phase_id, prize_item_id, tier, quantity

Winner
  draw_id, enrolment_id, prize_assignment_id, ticket_ref, status

Payout / Fulfilment
  winner_id, cbs_ref | handover_ref | scholarship_contract_ref

BehaviourEvent (optional normalized feed)
  customer_number, event_type, channel, amount, occurred_at, source_system

AuditEvent
  actor, action, entity, payload, timestamp
```

External keys: CBS `customer_number`, `account_number` (same pattern as DECSI Loan Hub integrations).

9. Integration contracts

9.1 Core banking (Temenos)

Capability	Pattern
Customer / account lookup	Sync API
Deposit / txn feed	Event webhook or near-real-time poll / batch (≤ 15 min latency target)
Balance history	For ADB campaigns
Prize cash credit	Idempotent payment API
Prize expense GL	Journal / designated marketing GL

Note (roadmap source): Phase 1 development states platform already developed/tested with DECSI Temenos core banking — integration hardening and production ops remain in scope for pilot.

9.2 AdoPay

Capability	Purpose
Deep link / in-app PLSA section	Show tickets, campaigns, T&Cs
Push notifications	Engagement
Channel flag on deposits	Digital ticket multiplier
Registration drives	Attribute new AdoPay users to PLSA campaigns

Pilot constraint: Phase 2 pilot is **AdoPay-only** participation path (see §14).

9.3 SMS gateway

OTP-free informational SMS; bulk campaign blasts; winner notices; complaint short-code optional.

9.4 Ecosystem (phase-gated)

System	Entry / promotion use
DECSI Loan Hub	Timely repayment events; cross-sell loan offers to savers
Seqela School ERP	Education campaign participation; scholarship school roster
DECSI SACCO / multipurpose	Group participation flags

9.5 Illustrative PLSA APIs

Method	Path	Description
POST	/api/v1/campaigns	Create campaign
POST	/api/v1/campaigns/{id}/phases	Add phase
POST	/api/v1/campaigns/{id}/approve	Checker approve
POST	/api/v1/enrolments	Opt-in
POST	/api/v1/events/transactions	CBS/AdoPay txn ingest
GET	/api/v1/enrolments/{id}/tickets	Customer/staff ticket view
POST	/api/v1/draws/{id}/freeze	Freeze census
POST	/api/v1/draws/{id}/execute	Dual-control draw
GET	/api/v1/draws/{id}/results	Results feed (SMS/radio/web)
POST	/api/v1/winners/{id}/verify	Branch verify
POST	/api/v1/winners/{id}/fulfil	Cash / in-kind / scholarship
GET	/api/v1/analytics/summary	M&E dashboard API

OpenAPI 3.0 required for all external integrations.

10. Staff roles and governance

Role	Location	Responsibilities
PLSA Program Management Team	HO	Campaign planning/approval, deposit monitoring, draws, marketing coordination
Campaign Maker / Checker	HO	Configure vs approve rules & prizes
Draw Committee	HO (+ auditor)	Authorize freeze/execute; witness
Compliance / Internal Audit	HO	Oversight, AML, draw audit
Branch PLSA focal person	Each branch	Promote, enrol, winner verification, local reporting
Finance	HO	Prize pool budget, GL, tax
Marketing	HO	Pillars, creative, media, community calendar

Aligns with marketing **implementation roadmap** and M&E governance in the DECSI proposal.

11. Customer journeys (MVP+)

11.1 Enrol (branch or AdoPay)

1. Identify customer in CBS
2. Select eligible savings account
3. Accept T&Cs (versioned)
4. Confirm via SMS / AdoPay
5. Start earning tickets on qualifying activity

11.2 Save and earn

1. Deposit via branch / agent / AdoPay
2. Integration layer notifies PLSA
3. Eligibility engine grants tickets
4. Optional SMS: "You now have N entries for {draw date}"

11.3 Draw and win

1. Freeze → dual-control draw → publish results
 2. Winner SMS + branch list + radio/social feed
 3. Claim window → KYC verify → fulfil
 4. Public celebration events for trust
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12. Compliance and risk controls

NBE-aligned pillars from *PLSA For DECSI* §10:

1. **Consumer protection** — published rules; no guaranteed-win marketing; complaint channel
2. **AML / KYC** — only verified CBS customers; monitor unusual deposit/withdrawal patterns; report as required
3. **Deposit protection** — full withdrawal rights; prize funds separated from customer deposits
4. **Fair, transparent, auditable draws** — RNG, independent oversight, public winners, immutable logs

Risks & mitigations: low participation (marketing pillars + pilot learnings); operational errors (automation + maker-checker); fraud (eligibility re-check, identity rules, audit).

13. Marketing pillars → system features

Pillar	Marketing intent	Platform feature
1 Savings mobilization	Save Today, Secure Tomorrow	Deposit tickets, seasonal campaigns, challenges
2 AdoPay adoption	“Your DECSI Account in Your Pocket”	Digital multipliers, in-app PLSA, registration attribution
3 Product awareness / cross-sell	Behaviour-based offers	Rule engine + Loan Hub / product nudges
4 Community & inclusion	Grassroots trust	Segment campaigns, geo targeting, SMS/radio result feeds
5 Loyalty / lottery rewards	Multi-behaviour entries	Unified ticket ledger + diverse prize catalogue + scholarships

Channels to support operationally: Facebook, Telegram, TikTok, YouTube, SMS, email, AdoPay, website, radio (Tigrigna), community meetings, OOH, print, branch activations, field agents, ambassadors, coop mobilizers, merchants.

14. Implementation roadmap

From DECSI proposal §13–14, refined for engineering delivery.

Phase	Scope	Notes
1 — System development	Campaign, eligibility, draw, prize, SMS, CBS integration	Source states already developed/tested with Temenos — complete UAT, harden, document APIs
2 — Pilot	AdoPay-only pilot cohort	Prove digital ticket path, notifications, ops playbooks
3 — Regional rollout	Mekelle branches (+ teller/agent if approved)	Add branch enrolment & in-kind fulfilment
4 — National deployment	All branches	Full channel set

Geographic rollout clusters

1. Urban centers incl. Addis Ababa
2. Major regional cities
3. Rural districts

Marketing timeline (parallel)

Marketing phase	Timing	Focus
Preparation	Month 1	Materials, training, stakeholder readiness
Launch	Months 2–4	Launch events, savings + AdoPay drives, referrals
Expansion	Months 5–12	Continuous campaigns, rewards, merchants, cross-sell
Optimization	Year 2+	Segmentation, personalization, new groups

15. KPIs and M&E (must be in analytics)

Savings

- Total savings growth; new accounts; average balance; voluntary participation rate

AdoPay

- Registrations; MAU; txn count/value; merchant enrolments; % PLSA participants using digital

Acquisition & engagement

- New customers; referral conversions; campaign participation; loyalty participation; retention

Product utilization

- Loan applications from PLSA cohort; cross-sold products; multi-product ratio

Ops / trust

- Draw complaint rate; payout STP %; ticket exception rate; branch league tables

Review cadence: monthly performance · quarterly impact · annual strategic evaluation

16. Non-functional requirements

Area	Target
Scale	≥ 1M customers; concurrent multi-campaign; peak harvest deposit bursts
Ticket/eligibility latency	≤ 15 minutes from CBS post (pilot); stretch to near-real-time
Availability	99.5% inquiry; planned draw windows
Security	TLS, RBAC, encrypted PII, secure RNG, audit retain ≥ 7 years
Localization	EN / AM / TI templates
DR	RPO ≤ 24h, RTO ≤ 8h (pilot); tighten for national

17. Open parameters (Board / Marketing lock)

1. Ticket unit (ETB) and max tickets per customer per draw
 2. Balance basis: net deposit vs ADB vs closing balance
 3. AdoPay multiplier value
 4. Which behaviours earn tickets in Phase-2 pilot vs national
 5. Prize mix: cash % vs in-kind % vs scholarship reserve
 6. Prize pool % (0.5 / 1 / 2) for year 1
 7. Employee / related-party exclusion rules
 8. Claim window days and unclaimed prize policy
 9. Scholarship partner school list and multi-year GL treatment
 10. Pilot success gates before Mekelle / national expansion
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18. MVP acceptance criteria

- ☐ Campaign → Phase → Draw configured with maker-checker
 - ☐ CBS deposit ingest creates auditable tickets
 - ☐ AdoPay-channel deposits receive configured multiplier
 - ☐ Freeze + dual-control draw + replayable audit pack
 - ☐ SMS notifications for enrolment, tickets, winners
 - ☐ Cash prize posting idempotent to CBS
 - ☐ Branch/HO dashboards for participation & deposit growth
 - ☐ Published T&Cs and responsible-marketing checklist signed off
 - ☐ AdoPay pilot cohort completed with KPI baseline report
-

19. Technology recommendation

Layer	Suggestion
PLSA services	Modular backend (Django/API or equivalent) + workers for eligibility/draws
DB	PostgreSQL
Integration	Secure APIs to Temenos, AdoPay, SMS; mocks for lower envs
Admin UI	HO campaign console + branch workbench
Customer UX	AdoPay module first; branch assisted enrolment
Analytics	Dashboard + exportable M&E packs
Deploy	Containerized; horizontal scale for national load

20. Document control

Version	Date	Notes
1.0	2026-07-28	Initial generic PLSA tech draft
2.0	2026-07-28	Rewritten from <i>PLSA For DECSI</i> + <i>PLSA Project Marketing Plan & Strategy (3)</i>

Next step: Freeze §17 parameters in a DECSI workshop; produce CBS/AdoPay OpenAPI bindings and pilot SoW.